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(a)(i)	Electrons are emitted from zinc when ultraviolet radiation is incident on it, discharging the gold leaf.
	Frequency of ultraviolet radiation, but not red light, exceeds threshold frequency of zinc.
(a)(ii)	Increasing intensity increases the number of photons incident per unit time.
	Energy of each photon doesn't change. Since electromagnetic radiation behave as particles which interact with electrons in a one-to-one manner, no electrons are emitted as energy of photon is still less than the work function of zinc.
(b)(i)	120 000 V
(b)(ii)	High energy electrons knock out the inner shell electrons from the metal target atoms. Electrons from higher energy level drops to lower energy level.
	The loss in energy of electrons is converted to X-ray photons of energy = energy difference between the two energy levels.
(b)(iii)	Different metal targets have their unique energy levels. Hence the differences are energy levels are unique.
(b)(iv)	Lower energy X-rays have insufficient energy to penetrate through the body. But they contribute to the overall dose in patients.
(c)(i)	Estimate uncertainty of position of $x = 24 \times 10^{-6} \text{ m}$
	$\Delta x \Delta p_x \geq h$ $\Delta p_x = \frac{6.63 \times 10^{-34}}{24 \times 10^{-6}}$ $= 2.8 \times 10^{-29} \text{ kg m s}^{-1} \text{ (round up to 2sf)}$

(c)(ii)	$p_y = m_e v = 1.0932 \times 10^{-27} \text{ kg m s}^{-1}$
	$\tan \theta = \frac{p_x}{p_y} = \frac{2.7625 \times 10^{-29}}{1.0932 \times 10^{-27}}$
	$\theta = 1.4^\circ \text{ (1dp)}$
	Alternative Wavelength associated with the electrons $\lambda = \frac{h}{p_y} = 6.065 \times 10^{-7} \text{ m}$
	Using single slit diffraction formula, $n = 1, b = 24 \times 10^{-6} \text{ m}$ $b \sin \theta = \lambda$
	$\theta = 1.448507^\circ = 1.4^\circ \text{ (1dp)}$
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